

8/10/2026 - RECURSIVE FEASIBILITY + STABILITY (INTRO).

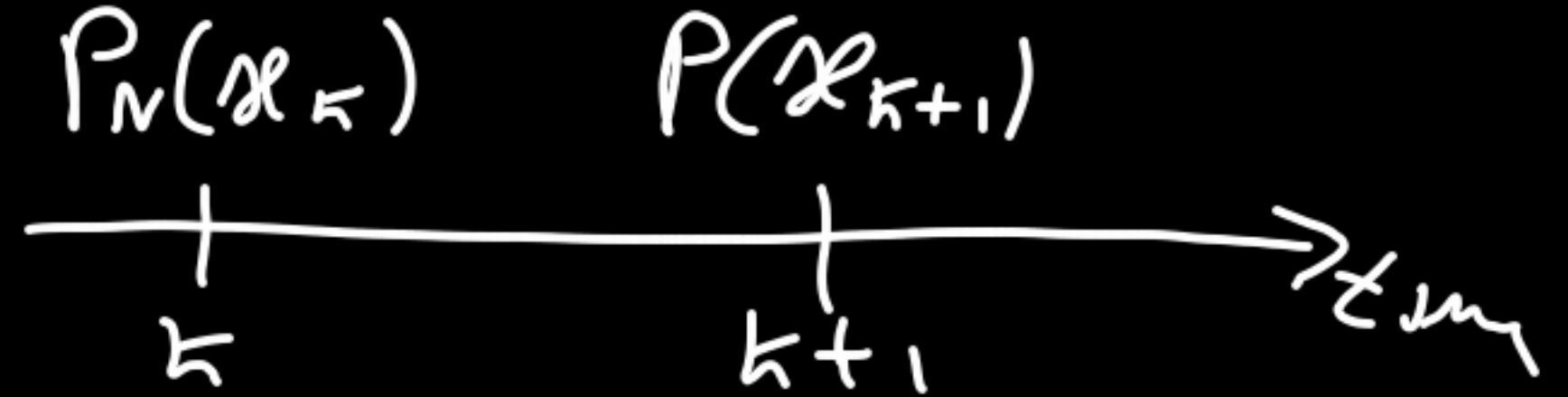
HOW TO PROVE REC. FEAS.

WE HAVE TO PROVE THAT:

$\forall x_k$: MPC ITERATION $P_N(x_k)$ IS FEASIBLE
($\forall x_k \in \mathcal{I}$), THE NEXT MPC ITERATION IS
FEASIBLE ($P_N(x_{k+1})$ HAS AT LEAST ONE FEASIBLE
SOLUTION).

CONSTRUCTIVE APPROACH:

WE USE THE ^{OPTIMAL} EVOLUTION OF $P_N(x_k)$ TO FIND A FEASIBLE SOLUTION FOR $P_N(x_{k+1})$.



ASSUMPTION (MAIN): NOMINAL MPC - MPC USES A PERFECT MODEL OF THE CONTROLLED SYSTEM.

$$x_{k+1} = f(x_k, u_{0|k}^*)$$

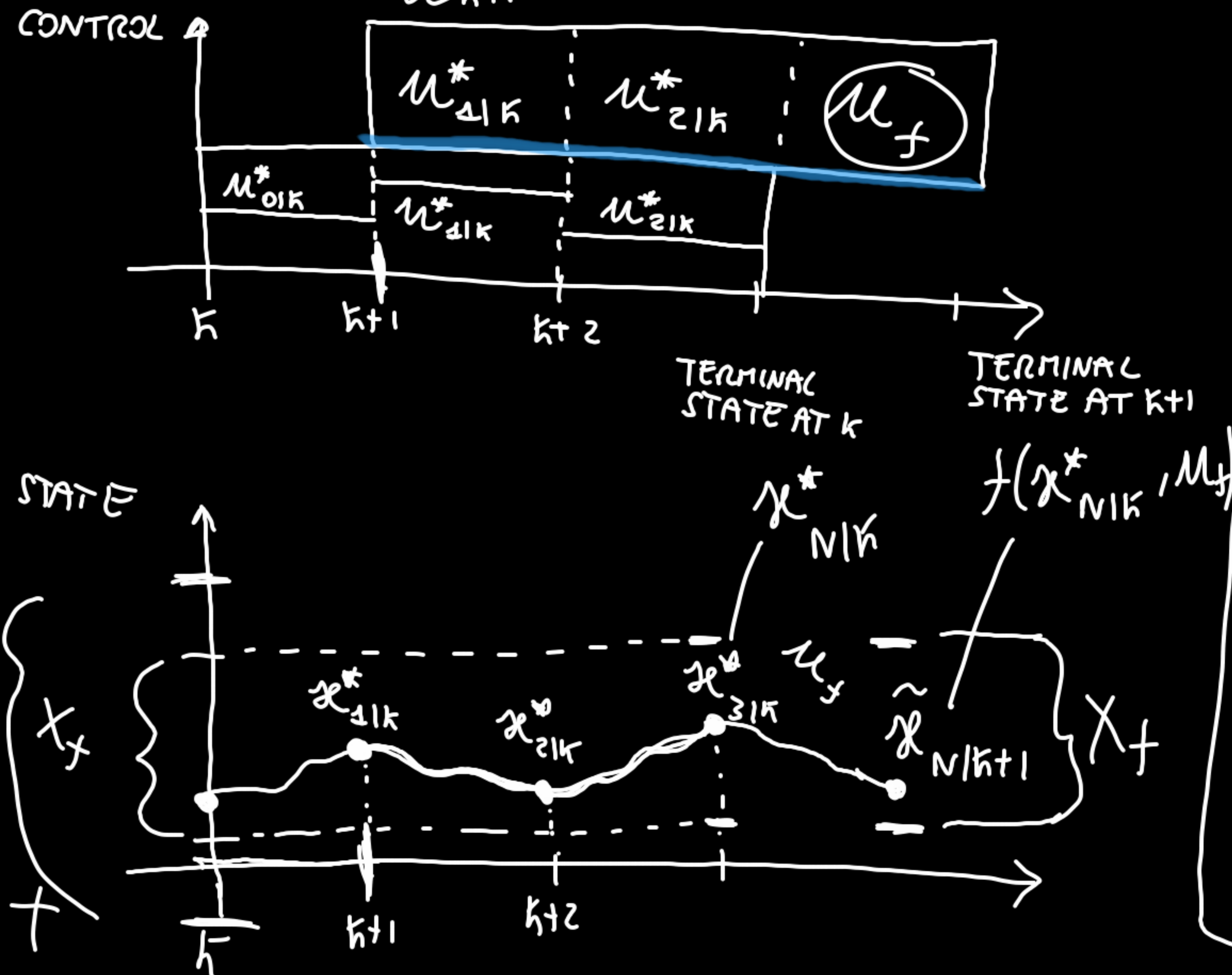
OPTIMAL SOLUTION OF $P_N(x_k)$, FOR A GENERIC $x_k \in \mathcal{F}$.

$$u_k^* = \{u_{0|k}^*, u_{1|k}^*, \dots, u_{N-1|k}^*\}$$

$$x_k^* = \left\{ \begin{array}{c} x_{0|k}^*, x_{1|k}^*, \dots, x_{N-1|k}^*, x_{N|k}^* \\ ||| \\ x_k \end{array} \right\}$$

VISUALIZATION OF OPTIMAL SOLUTION OF $P_N(x_k)$ ($N=3$)

$\tilde{u}_{k+1}$ TENTATIVE



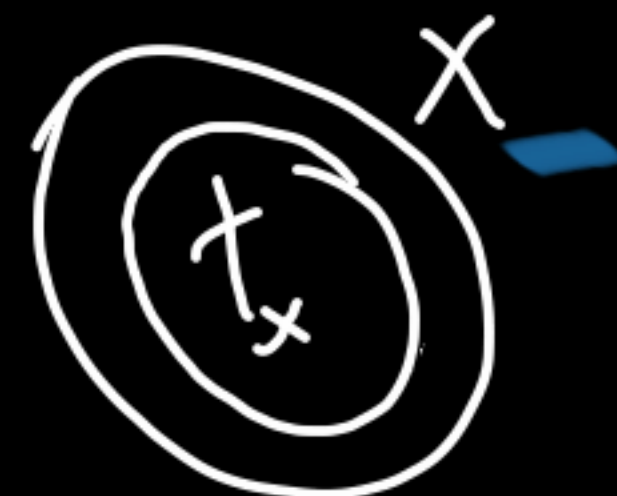
$$\min \left\{ \sum_{i=0}^{N-1} \ell(x_{i|k}, u_{i|k}) + V_f(x_{N|k}) \right\}$$

s.t.

$$\begin{cases} x_{i+1|k} = f(x_{i|k}, u_{i|k}), i=0,1,\dots,N-1 \\ x_{0|k} = x_k \end{cases}$$

$$\begin{cases} u_{i|k} \in U, i=0,1,\dots,N-1 \\ x_{i|k} \in X, i=0,1,\dots,N \end{cases}$$

$$x_{N|k} \in X_f \text{ (TERMINAL CONSTR.)}$$



$$u_k^* = \left\{ \underbrace{u_{0k}^*}_{\text{WE APPLY THIS AT } k}, \overbrace{u_{1k}^*, \dots, u_{Nk}^*}^{\text{TAIL}} \right\}$$

WE APPLY
THIS AT
k.

$$\tilde{u}_{k+1} = \left\{ \underbrace{u_{1k}^*, \dots, u_{Nk}^*}_{\text{TAIL OF OPTIMAL SOL. AT } k}, \underbrace{u^+}_{\text{FEASIBLE TERMINAL CONTROL}} \right\}$$

WE HAVE TO FIND (IF POSSIBLE), A
TERMINAL CONTROL THAT MAKES
 $\tilde{u}_{k+1}$ FEASIBLE FOR $P_N(x_{k+1})$.

WE HAVE TO CHECK THIS:

- $u_f \in U$ (WE USE
A FEASIBLE TERMINAL
CONTROL.

- THE TERMINAL STATE
OF ITERATION AT TIME
k+1 IS IN THE TERMINAL
SET

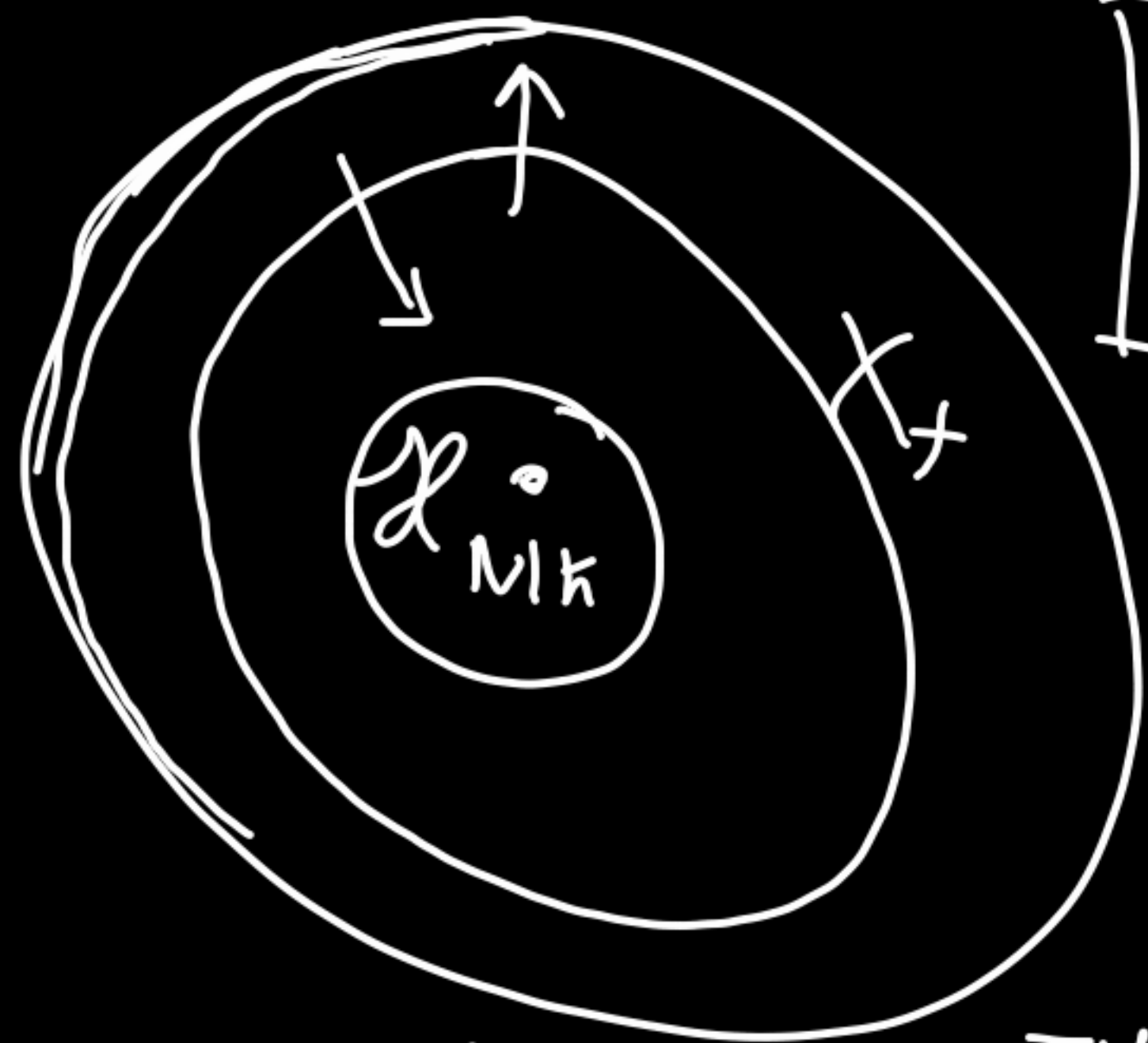
$$x_{N|k+1} = f(x_{N|k}, u_f)$$

IF I CAN FIND A TERMINAL CONTROL LAW u_f AND
A TERMINAL SET X_f SUCH THAT u_f MAKES
 X_f CONTROLLED POSITIVE INVARIANT W.R.T

THE CONTROLLED SYSTEM $x_{k+1} = f(x_k, u_k)$, THEN I
AM SURE THAT THE TENTATIVE SOLUTION

$\tilde{u}_{k+1} = \{u_{1k}^*, u_{2k}^*, \dots, u_{N-1k}^*, u^f\}$ IS FEASIBLE FOR $P(x_{k+1})$.

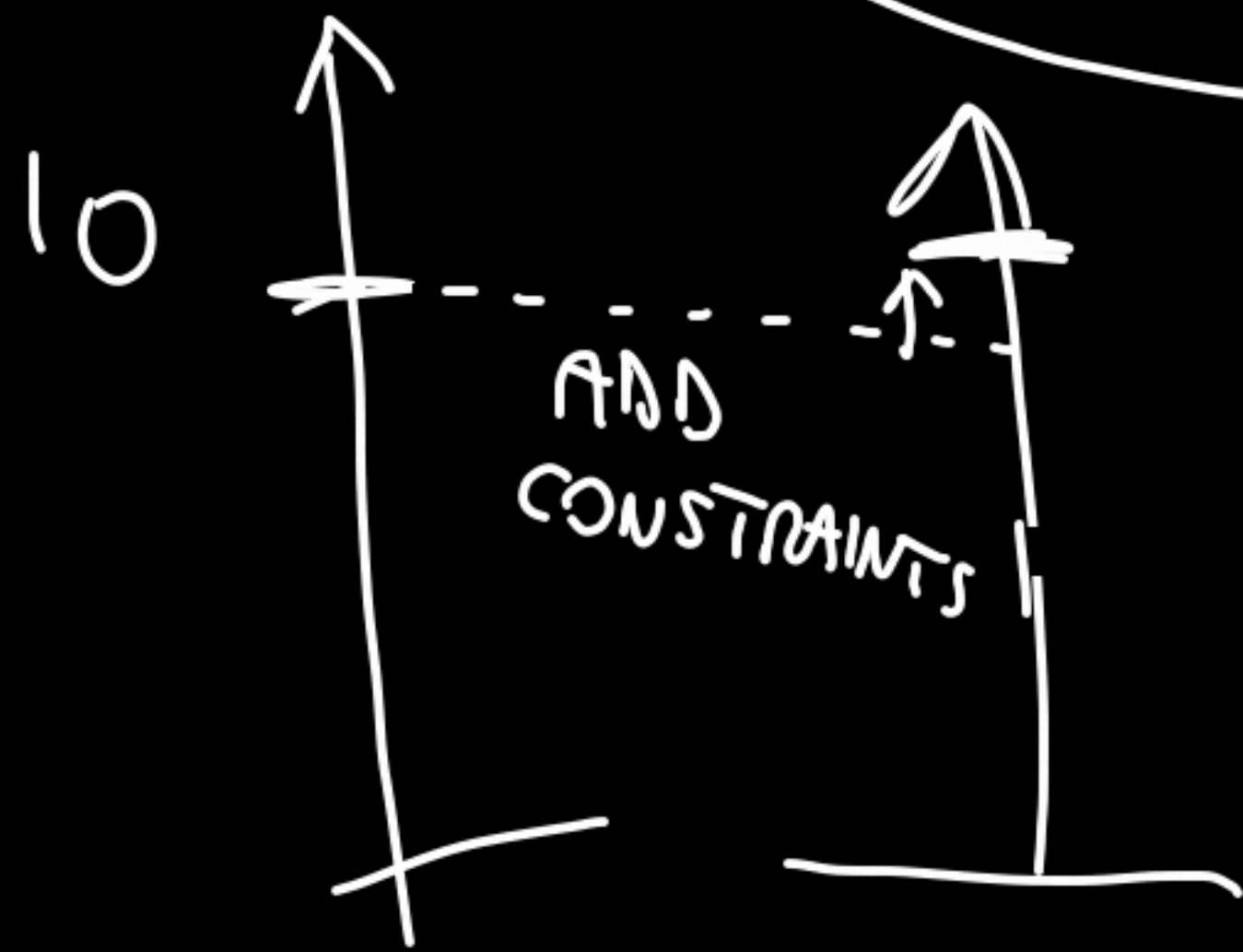
C.P.I. OF X_f WRT $f \iff \forall x \in X_f \exists u : f(x, u) \in X_f$.



$$X = X_f$$

u_f

OPTIMAL
COST

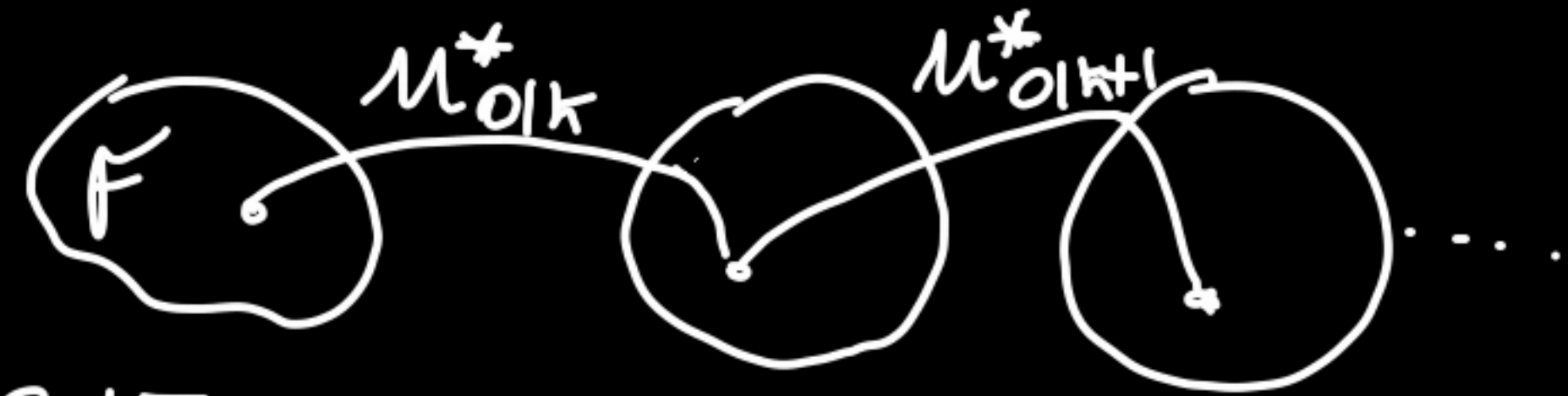


WE SHOULD SEARCH FOR

THE LARGEST POSSIBLE X_f
FOR WHICH WE CAN FIND
A TERMINAL CONTROL u_f THAT
MAKES $IT(X_f)$ C.P.I. W.R.T
THE SYSTEM $f()$.

RECURSIVE FEASIBILITY $\iff$ (THE MPC ALGORITHM) IT MAKES $\mathcal{F}$ C.P.I. WRT THE CONTROLLED SYSTEM.

$\forall x_k \in \mathcal{F} \rightarrow$ I CAN FIND A FEASIBLE CONTROL SEQUENCE FOR THE MPC ITERATION AT $k+1$.



(TAIL-BASED APPROACH + NOMINAL ASSUMPTION) (N VECTORS OF CONTROL)

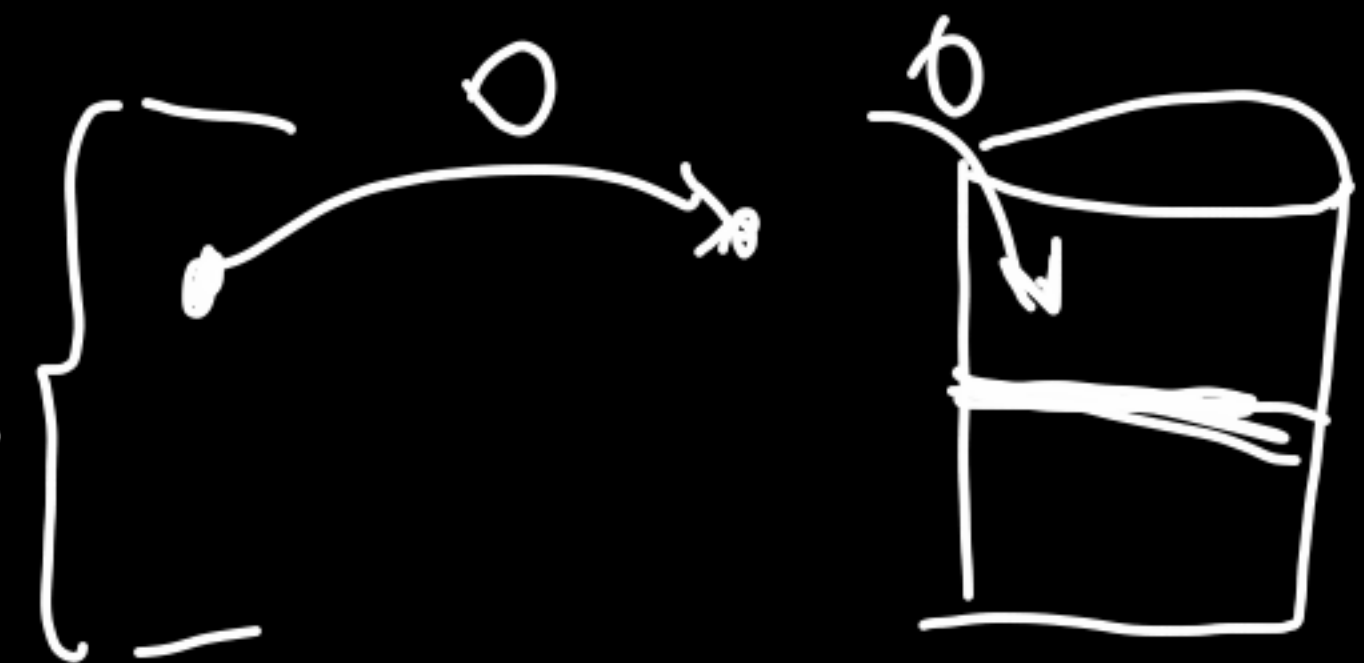
- FIND x_f AND $\underline{u}_f$ SUCH THAT $\underline{u}_f$ MAKES x_f C.P.I. WRT f

FOR EXAMPLE:

- $\bullet u_f = 0$

(DO NOTHING) $x_f = x$

- $\bullet u_f = K x_{N|k}$ (LINEAR CONTROL)



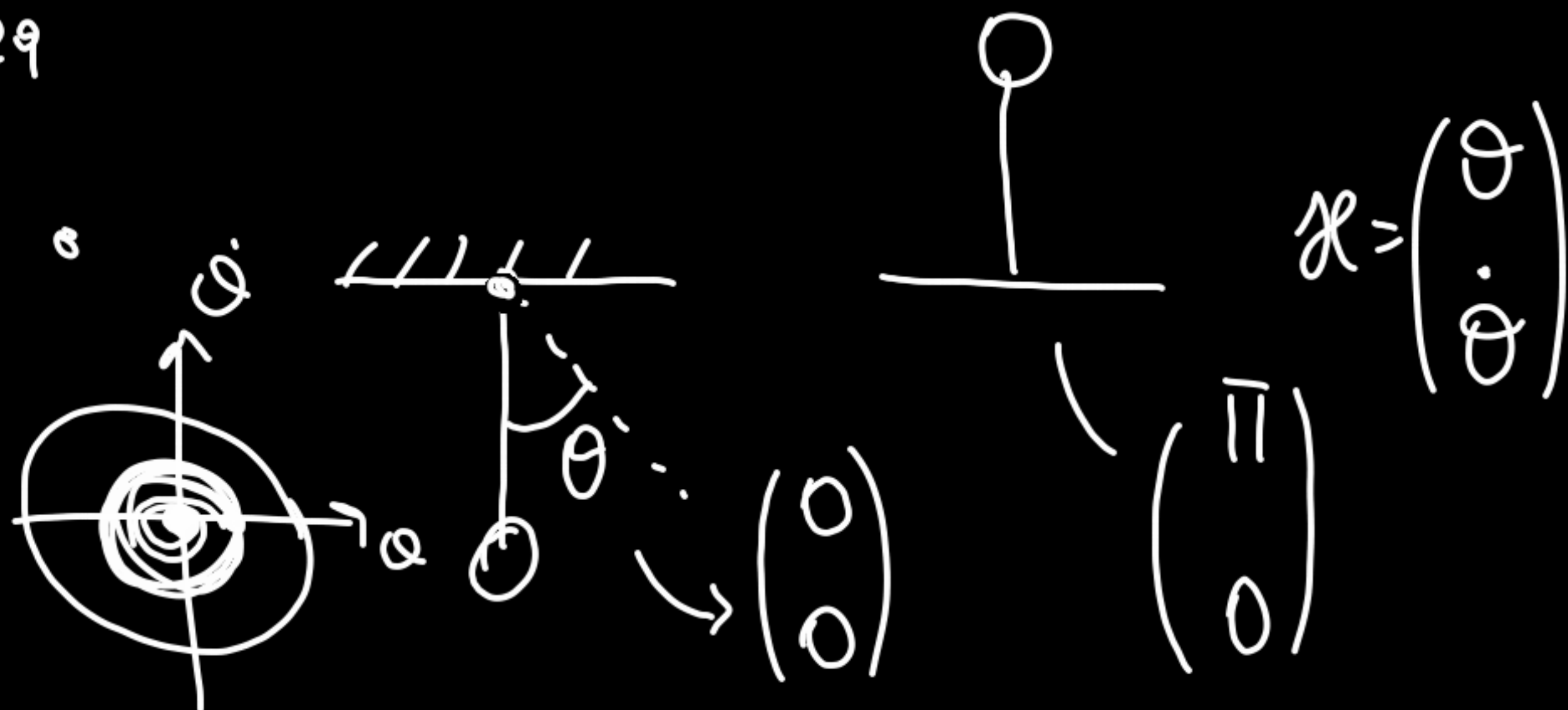
STABILITY OF MPC

STABILITY IS A PROPERTY OF THE EQUILIBRIUM POINTS OF A DYNAMICAL SYSTEM $x_{k+1} = f(x_k, u_k)$.

(UNFORCED) EQUILIBRIUM POINT:

A STATE x^{eq} IS AN EQUILIBRIUM FOR THE SYSTEM

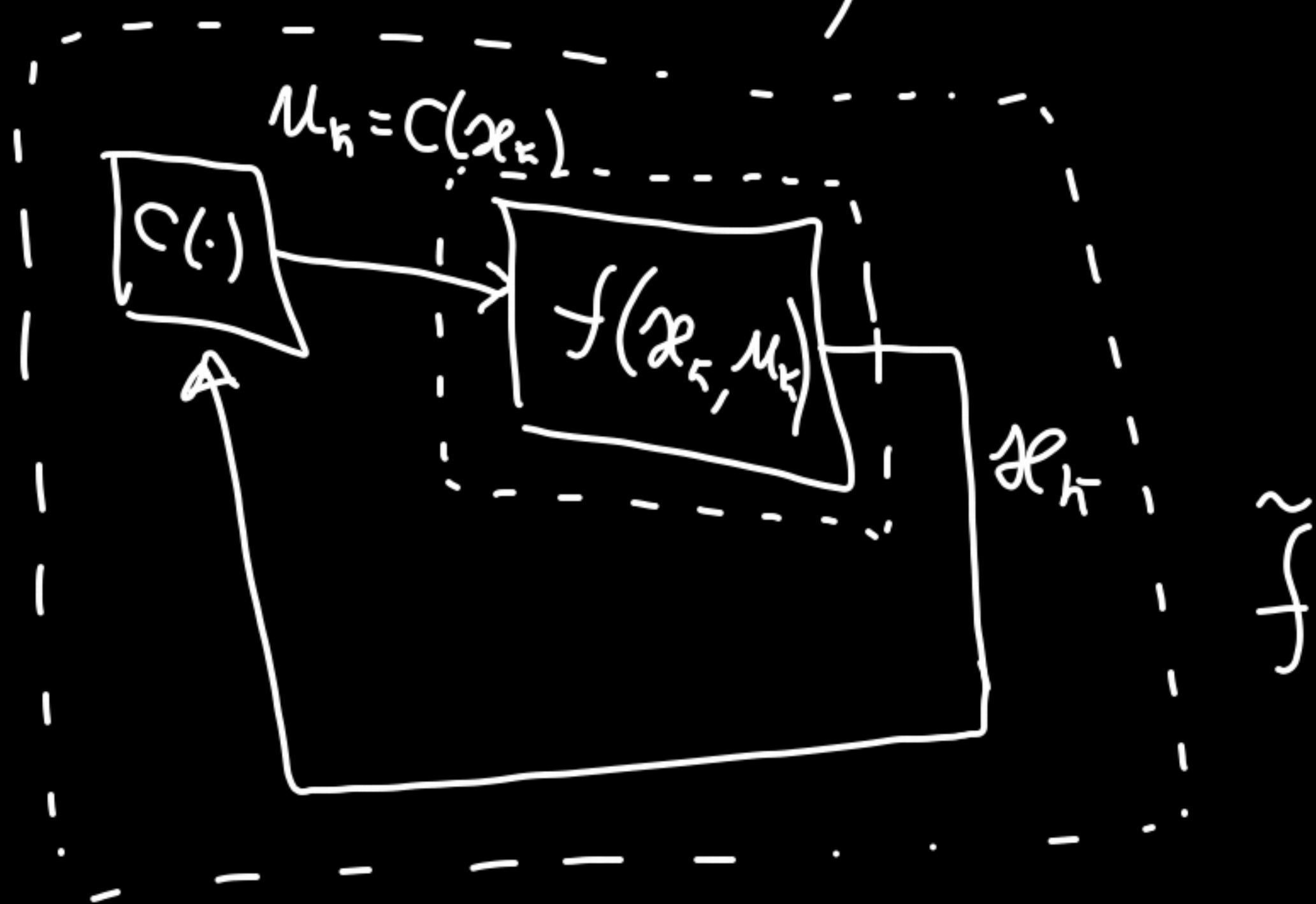
IFF $f(x^{eq}, 0) = x^{eq}$



A SYSTEM IN GENERAL CAN HAVE NO, ONE, FEW,
MANY, INFINITE-MANY EQ. POINTS.

STABILIZATION OF A SYSTEM

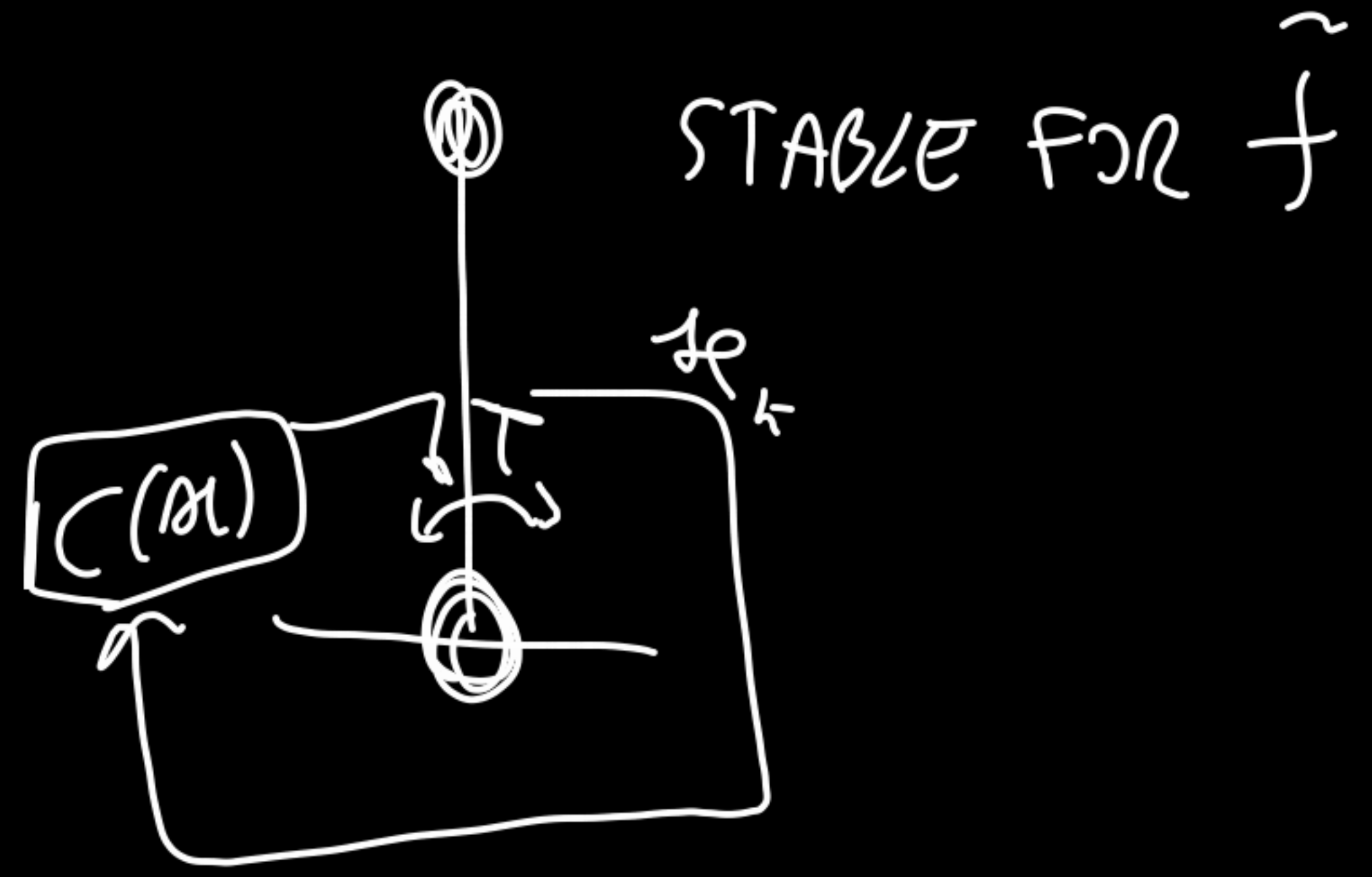
$x_{k+1} = f(x_k, u_k)$, CONSIDER A FEEDBACK LAW $u_k = C(x_k)$



$$x_{k+1} = f(x_k, C(x_k)) = \tilde{f}(x_k)$$

TYPICAL PROBLEMS

- STABILIZATION: FIND A CONTROL LAW $C(x_k)$ THAT MAKES AN UNSTABLE EQ. POINT OF f A STABLE EQ. POINT FOR $\tilde{f}$.



— EVEN WHEN x^{eq} OF f IS STABLE, WE MUST CHECK THAT IT REMAINS STABLE ALSO FOR $\tilde{f}$ (I.E., AFTER WE ADD CONTROL).

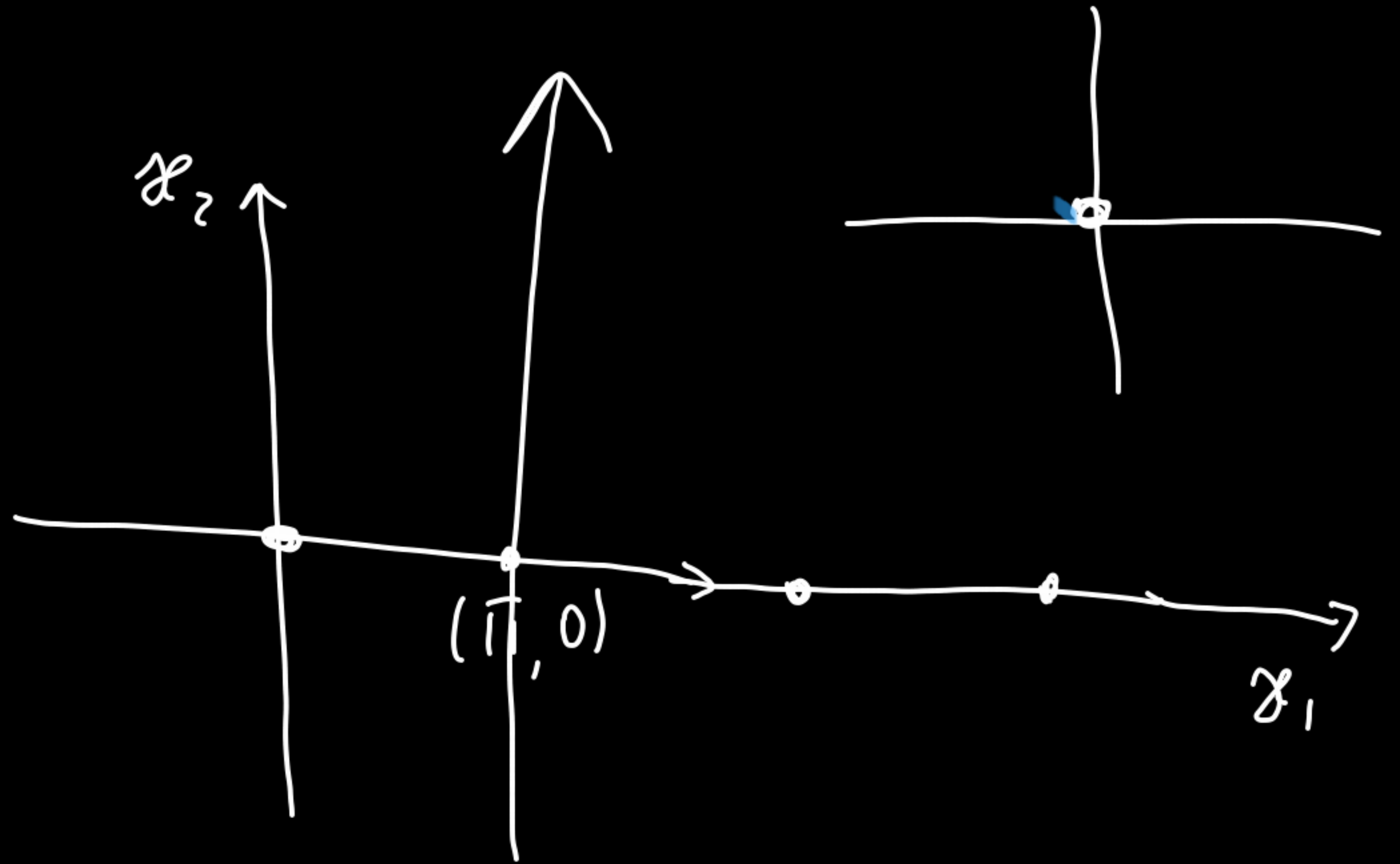


$$x_{k+1} = f(x_k)$$

WE WORK WITH THIS EQUATION

WITH NO LOSS OF GENERALITY, WE ALWAYS CONSIDER

$$x^{\text{eq}} = 0.$$



DEFINITION OF STABILITY, ASYMPTOTIC STABILITY, UNSTABILITY

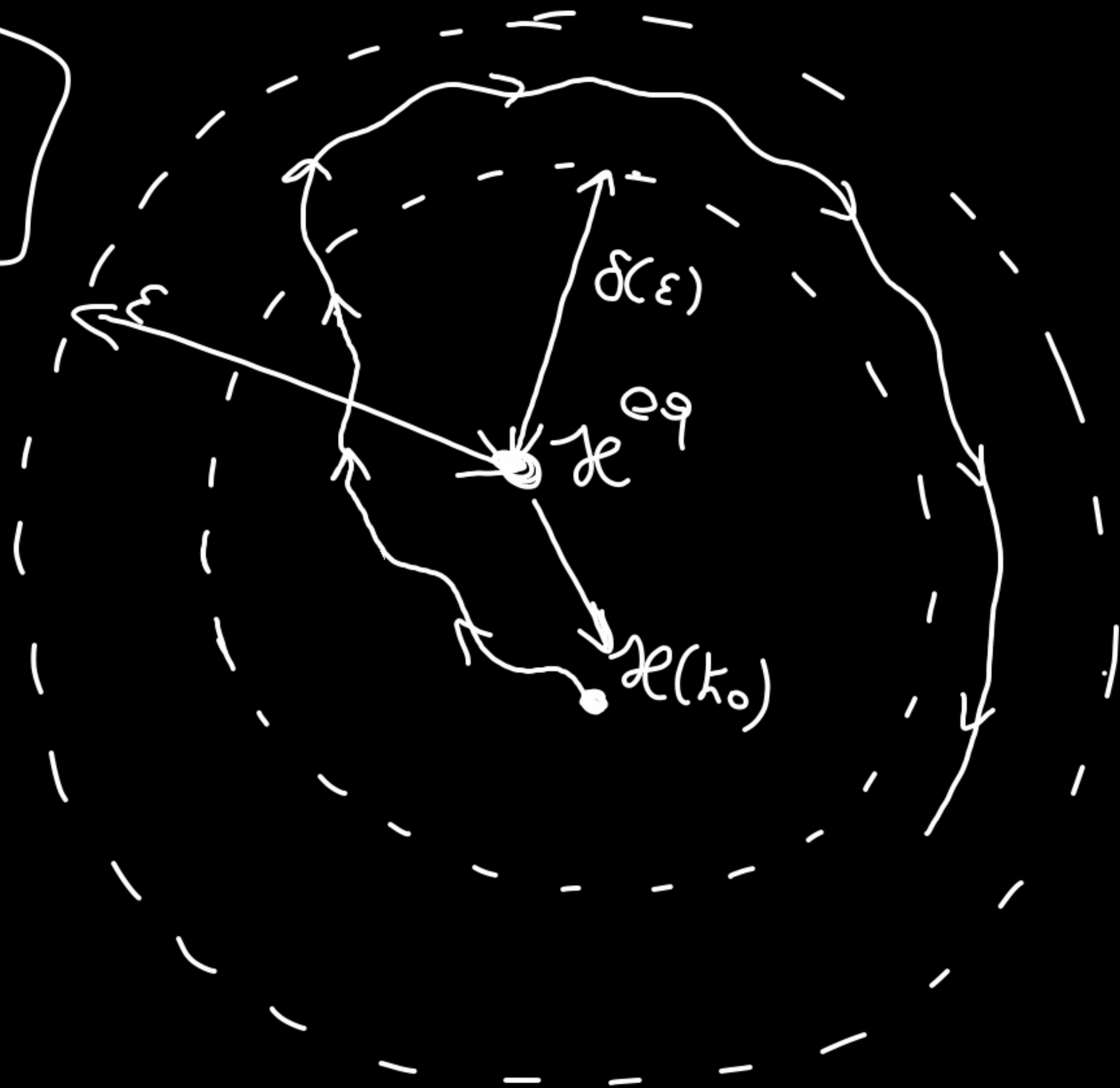
DEF. (STABLE)^(s) x^{eq} . AN EQ. POINT x^{eq} OF SYSTEM $x_{k+1} = f(x_k)$ IS STABLE IFF:

ON $\left[\begin{array}{l} \text{SMALL INITIAL PERTURBATIONS OF THE EQUILIBRIUM WILL} \\ \text{RESULT IN STATE TRAJECTORIES THAT WILL REMAIN} \\ \text{FOREVER CLOSE TO } x^{eq}. \end{array} \right.$

LOCALLY: $\left[\begin{array}{l} \forall \varepsilon > 0, \exists \delta(\varepsilon) > 0 : \|x(k_0) - x^{eq}\| \leq \delta(\varepsilon) \Rightarrow \|x(k) - x^{eq}\| \leq \varepsilon, \forall k \geq k_0. \\ \delta(\varepsilon) \leq \varepsilon \end{array} \right.$

PICTURE FOR STABLE x^{eq} (S)

LOCAL
CONCEPT

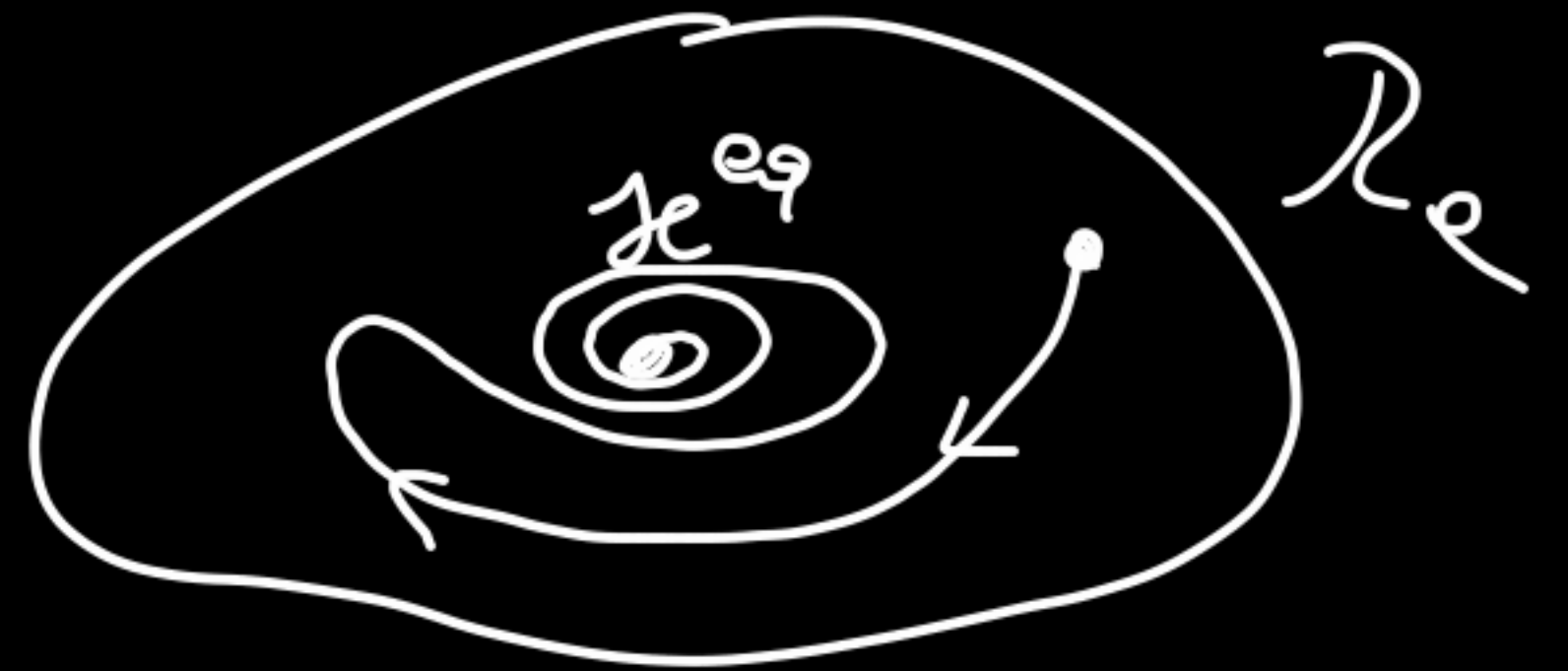


$x^{eq,2}$

ASYMPTOTIC STABILITY: (AS)

AN EQ. POINT x^{eq} IS (LOCALLY) ASYMPTOTICALLY STABLE IFF:

① IT IS STABLE;



② (ATTRACTIVENESS): $\exists R_a \subseteq X$ (THERE EXIST

A REGION OF ATTRACTION: IF $x(k_0) \in R_a \Rightarrow$

$\rightarrow x(k) \rightarrow x^{eq}$.

UNSTABLE x^{eq}

- x^{eq} THAT IS NOT STABLE

